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(b) A and B locked together after collision.

(i) Find the speed of them just after collision. (2 marks)

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(ii) If the collision time between the trains is 0.2 s and the mass of each train is 5000 kg , find the magnitude and direction of the average impact force acted on A during collision. (3 marks)

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5. (a) A bow and arrow is a kind of projectile weapon. The string of a bow is drawn taut by a hunter with a force of 60 N and an arrow of mass 0.2 kg is held stationary as shown in Figure 5.1.

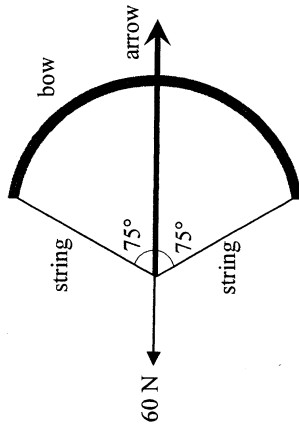


Figure 5.1

- (i) Find the tension of the string. Neglect the weight of the arrow. (2 marks)

- (ii) Estimate the energy stored in the taut string if the initial speed of the arrow is 45 m s^{-1} when released. Assume that the bow is rigid and neglect the mass of the string. (2 marks)

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2.



Figure 2.1

In a road test, John drives his car along a straight horizontal road (see Figure 2.1). The car takes 9.3 s to accelerate from rest to 100 km h^{-1} . The total mass of John and his car is 1400 kg.
(Note: $100 \text{ km h}^{-1} = 27.8 \text{ m s}^{-1}$)

- (a) Find the total kinetic energy of John and his car when travelling at 100 km h^{-1} . Hence estimate the average output power of the car during this acceleration. (3 marks)

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- (b) A similar road test is conducted on an inclined road. The car now takes more than 9.3 s to accelerate from rest to 100 km h^{-1} along the road (see Figure 2.2). Assume the output power of the car remains unchanged. Explain why it takes a longer time for the car to accelerate up an inclined road than along a horizontal road. (2 marks)

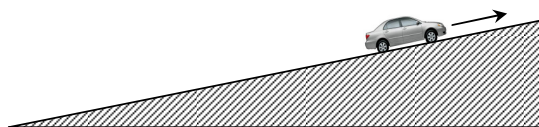


Figure 2.2

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